PSTAT 188: What is Data Science

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Week 2a: Intro to Data Science

Data Science Lifecycle

Working with tabular data

Data visualization

Python Jupyter notebooks using *pandas* and *seaborn*

Goal is to be hands-on

Much of the material is borrowed/distilled from PSTAT 100 notes courtesy of Trevor Ruiz

What is data science?

Data science encompasses a wide range of activities that involve *uncovering insights from quantitative information*.

People that refer to themselves as **data scientists** typically combine specific interests (“domain knowledge”, *e.g.*, biology) with computation, mathematics, and statistics and probability to contribute to knowledge in their communities.

- Intersectional in nature
- No singular disciplinary background among practitioners

Data science lifecycle

> Data science **lifecycle**: *an end-to-end process resulting in a data analysis product*

- Question formulation
- Data collection and cleaning
- Exploration
- Analysis

These form a *cycle* in the sense that the steps are iterated for question refinement and futher discovery.

Data science lifecycle

The point isn't really the exact steps, but rather the notion of an iterative process.

> **-> Pandas basics in Google Colab/Jupyter**

Starting with a question

The scaling of brains with bodies is thought to contain clues about evolutionary patterns pertaining to intelligence.

There are lots of datasets out there with brain and body weight measurements, so let’s consider the question:

> *What is the relationship between an animal’s brain and body weight?*

Data acquisition

Allison *et al.* 1976 provide average body and brain weights for 62 mammals:

	species	body_wt	brain_wt
0	Africanelephant	6654.000	5712.0
1	Africangiantpouchedrat	1.000	6.6
2	ArcticFox	3.385	44.5

Units of measurement

- body weight in kilograms
- brain weight in grams

Data assessment

How well-matched is the data to our question?

- Mammals only (no birds, fish, reptiles, etc.)
- Species are those for which convenient specimens were available
- Averages across specimens are reported (‘aggregated’ data)

Data assessment

Based on the great points you just made, we really only stand to learn something about this **particular** sample of animals.

In other words, no *inference* is possible.

> Do you think the data are still useful?

Inspection

This dataset is already impeccably neat: each row is an observation for some species of mammal, and the columns are the two variables (average weight).

So no tidying needed – we'll just check the dimensions and see if any values are missing.

```
# dimensions?
bb_weights.shape
```

```
(62, 3)
```

```
# missing values?
bb_weights.isna().sum(axis = 0)
```

```
species     0
body_wt     0
brain_wt    0
dtype: int64
```

Exploration

Visualization is usually a good starting point for exploring data.

The few outliers skew the visuals so that we cannot discern the big clump around $(0, 0)$ – that suggests we shouldn't look for a relationship on the scale of kg/g.

Exploration

A simple **log-log transformation** of the axes reveals a clearer pattern.

Analysis

The plot shows us that there's a roughly **linear** relationship on the log scale:

$$\log(\text{brain}) = \alpha \log(\text{body}) + c$$

So what does that mean in terms of brain and body weights? A little algebra gives us a "power law":

$$(\text{brain}) \propto (\text{body})^{\alpha}$$

Check your understanding: what's the proportionality constant?

Interpretation

So it appears that the brain-body scaling is well-described by a **power law**:

> among selected specimens of these 62 species of mammal, species average brain weight is approximately proportional to a power of species average body weight

Compare to poorly worded versions like:

- animals' brains are proportional to a power of their bodies
- among these 62 mammals, average brain weight is approximately proportional to a power of average body weight [the average is per species not across all mammals]

Question **refinement**: we can now ask further, more specific questions:

> Do other types of animals exhibit the same power law relationship?

More data acquisition

To investigate, we need richer data.

A number of authors have compiled and published 'meta-analysis' datasets by combining the results of multiple studies.

Below we import a few of these for three different animal classes.

```
# import metaanalysis datasets
reptiles = pd.read_csv('../../datasets/reptile_meta.csv')
birds = pd.read_csv('../../datasets/bird_meta.csv', encoding = 'latin1')
mammals = pd.read_csv('../../datasets/mammal_meta.csv', encoding = 'latin1')
```

Data assessment

Where does this data come from? It's kind of a convenience sample of scientific data:

- Multiple studies $\rightarrow$ possibly different sampling and measurement protocols
- Criteria for inclusion unknown $\rightarrow$ probably neither comprehensive nor representative of all such measurements taken

So these data, while richer, are still relatively narrow in terms of generalizability.

Cannot do general inferences (*e.g.*, to all animals, all mammals, etc.) because they weren't collected for the purpose to which we're putting them.

Usually, if data are not collected for the explicit purpose of the question you're trying to answer, they won't constitute a **representative** sample.

Inspection

This dataset has quite a lot of **missing** brain weight measurements: many of the underlying studies combined to form these datasets did not include that particular measurement.

```
# missing values?
data.isna().mean(axis = 0)
```

```
Order      0.00000
Family     0.00000
Genus      0.00000
Species    0.00000
Sex        0.00000
body       0.00000
brain      0.57404
class      0.00000
dtype: float64
```

Exploration

Focusing on the non-missing values, we see the same power law relationship but with *different* proportionality constants and exponents for the three classes of animals.

So we might hypothesize that:

$$
\begin{aligned}
(\text{brain}) &= \beta_1 (\text{body})^{\alpha_1} & & (\text{mammal}) \\
(\text{brain}) &= \beta_2 (\text{body})^{\alpha_2} & & (\text{reptile }) \\
(\text{brain}) &= \beta_3 (\text{body})^{\alpha_3} & & (\text{bird })
\end{aligned}
$$

Interpretation

It seems that the average brain and body weights of the birds, mammals, and reptiles measured in these studies exhibit distinct power law relationships.

$$\beta_i \neq \beta_j, \quad \alpha_i \neq \alpha_j \quad \text{for } i \neq j$$

What would you be interested in investigating next?

- Correlates of body weight?
- Adjust for lifespan, habitat, predation, etc.?
- Estimate the α_i's and β_i's?
- Predict brain weights for unobserved species?

A comment

Notice that the word 'model' is never mentioned

This was intentional — it is a common misconception that analyzing data always involves fitting models.

- Models are not not always necessary or appropriate
- You can learn a lot from exploratory techniques
- Models approximate specific kinds of relationships in data
- Exploratory analysis can reveal unexpected structure

Zooming out: the Data Lifecycle

This example illustrates the aspects of the lifecycle:

- data retrieval and import
- tidying and transformation
- visualization
- exploratory analysis
- modeling

History of Data Science

- Data science emerged as a term of art in the last decade
- Interest exploded in the last five years

Origins: ‘data analysis’: In 1960s Tukey advocated for ‘data analysis’ as a broader field than statistics:

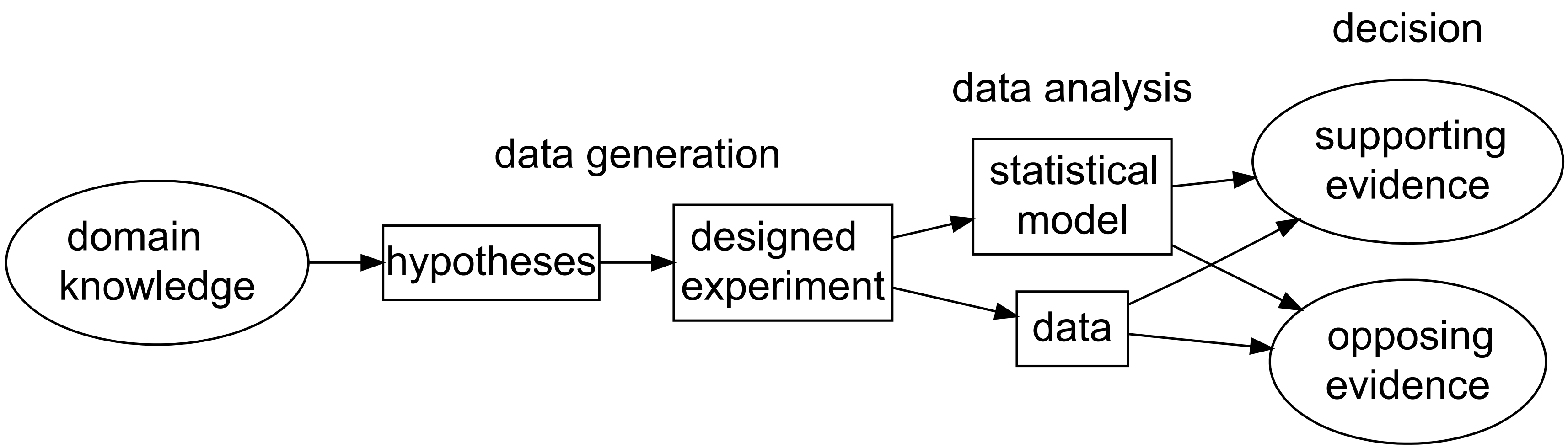
- statistical theory and methodology;
- visualization and data display techniques;
- computation and scalability;
- breadth of application.

Tukey’s ‘data analysis’ is proto-modern data science. In the 1960’s and 1970’s:

- visualization meant ***drawing***
- computation meant ***data re-expression by hand***

Confirmatory approach

The “confirmatory” approach of the classical inferential framework.



- statistical model determined by experimental design
- analysis based on statistical theory.
- output is a decision

Early data analysis concepts

The new techniques allowed for ***iterative*** investigation:

- formulate a question
- examine data graphics and summaries
- adjust computations and graphics to hone in on content of interest
- refine the question

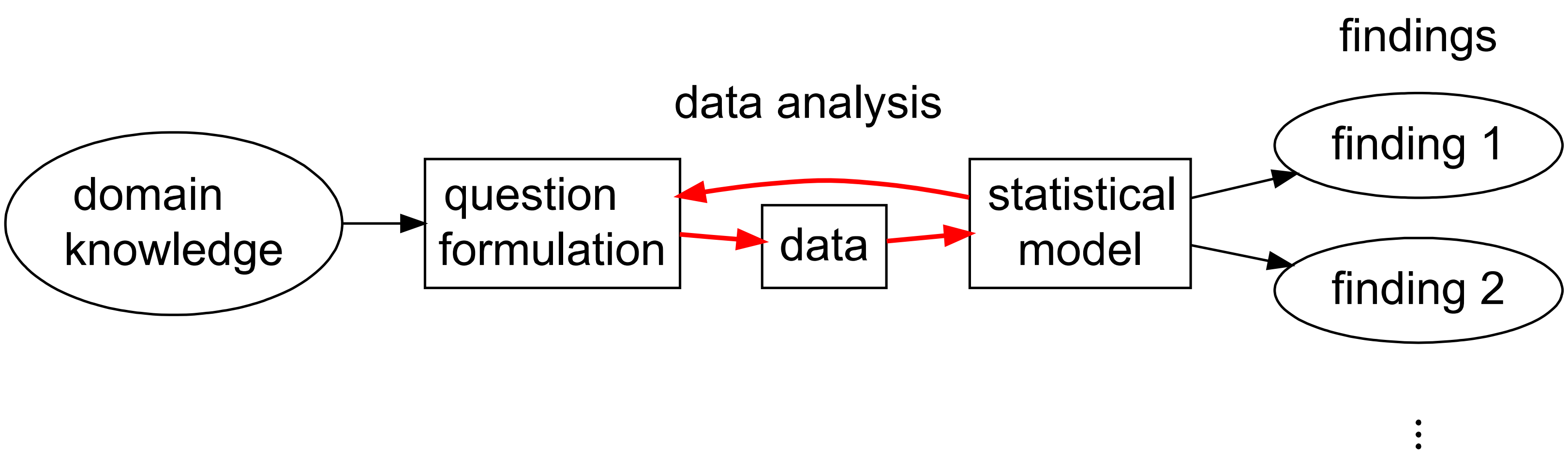
From 1960-2010, adopters of the 'data analysis as a field' view were largely industry practitioners. Their ideas evolved into an alternative approach:

- data-driven rather than theory-driven;
- iterative rather than conclusive.
- In the meantime, computer scientists advanced the field of machine learning

Exploratory approach

Iterative problem-solving together with applied machine learning was codified into an academic discipline in the early 21st century.

The “exploratory” approach of iterative modern data analysis.



- outputs are findings
- statistical model determined by data
- analysis techniques include empirical methods

Drivers of change

In the 2000s and especially after 2010, the iterative approach enjoys broader applicability than it used to:

- due to automated and/or scalable data collection
  - observational data is widely available across domains
  - and includes large numbers of variables
- highly specialized data problems evade methodology with theoretical support
- more accessible to analysts without advanced statistical training
- Machine learning advanced modern **predictive modeling**, most notably in the past 15 years:
  - neural networks and deep learning
  - optimization
  - algorithmic analysis

Structuring Data

Usually work with **tabular data**: Many ways to structure a dataset according to

1. Rows
2. Columns
3. Number of tables

Because of the wide range of possible layouts for a dataset, and the variety of choices that are made about how to store data, **data scientists are constantly faced with determining how best to reorganize datasets in a way that facilitates exploration and analysis.**

Next class: **tidy data format** and basics of **dataframes** in pandas